

Cross-Border Transactions: CS V3 Evaluation & Cloud-First Scoping Plan

Strategic intent

Command Cloud should **not replicate** the CS V3 on-prem cross-border model. CS V3 spreads currency, tax, and pricing concerns across implicit configuration (company codes, plant codes, tax codes, accounting categories) with no first-class currency model — workable for single-currency deployments but **disjointed and admin-heavy** for international operations.

Design goals for Command Cloud:

- 1. **Simplify setup** — admins configure a small number of tenant-level settings; the platform infers tax jurisdiction and currency defaults from delivery address, legal entity, and customer contract
- 2. **Multi-currency from MVP** — currency fields are mandatory in the data model from day one, even when transaction and functional currency are the same (EUR/EUR)
- 3. **Progressive enhancement** — MVP foundation → corridor compliance rules → multi-entity → advanced FX/treasury
- 4. **Do not carry forward CS V3 UX** — no "Pricing Company Code" / "Shipping Company Code" exposed to users; map these to cloud-native **Legal Entity** and **Location** models internally

```
flowchart LR
    subgraph legacy [CS V3 On-Prem]
        PC["Pricing Company Code"]
        SC["Shipping Company Code"]
        TC["Tax Code manual pick"]
        ImplicitCurr["Currency implicit"]
    end

    subgraph cloud [Command Cloud Target]
        Tenant["Tenant: region + currencies"]
        Entity["Legal Entity + VAT ID"]
        AutoTax["Auto tax from address"]
        ExplicitCurr["Currency on every record"]
    end

    legacy -.->|"import map only"| cloud
```

Investigation Summary (Current Evidence)

What we could and could not verify

Source	Status
CS V3 application source code	Not in workspace — /workspace contains only a README
Internal Confluence/Jira	Not accessible — Atlassian MCP requires authentication

Prototype docs on remote branches	Available — EU regional-gating pattern in docs/command-series-v3-hauler-license-expiry.md
Public Command Alkon product docs	Available — Sales & Quoting, Dispatch, Billing, APIs, Payments
Legacy CS user guides (COMMANDexecutive, customer training PDFs)	Available — pricing/tax hierarchy, no explicit currency fields

Important nuance for your France → Germany example: both countries use **EUR**. Cross-border complexity here is primarily **tax jurisdiction, VAT treatment, legal entity, and e-invoicing** — not FX conversion. Currency display will typically show **€** throughout; the meaningful differences are tax codes, company/plant routing, and invoice compliance metadata.

Northern Ireland / UK ↔ Republic of Ireland is a different class of problem: this corridor involves **two currencies (GBP and EUR), post-Brexit dual VAT rules** under the Windsor Framework, and an **XI-prefixed VAT identifier** for NI businesses trading goods with the EU. For International RMC customers operating border plants (common on the island of Ireland), this is likely the **highest-priority cross-border scenario** — and the one where multi-currency and FX support matter most.

Reference corridor comparison

Dimension	France → Germany	NI/UK ↔ Republic of Ireland
Currencies	EUR only	GBP (£) and EUR (€)
Primary complexity	Intra-EU VAT, e-invoicing	FX conversion + dual VAT regime + post-Brexit rules
VAT on goods (B2B)	Intra-EU reverse charge / zero-rated	NI ↔ ROI treated as intra-EU for goods (Windsor Framework); GB ↔ ROI is export/import
VAT on services	EU reverse charge	UK VAT rules apply (not EU goods rules)
VAT ID format	FR... / DE...	XI... prefix for NI; IE... for ROI
E-invoicing timeline	FR Sep 2026; DE already active	ROI Nov 2028 (phased); UK Apr 2029 (mandatory)
Command Alkon precedent	General EU regional gating	Hauler spec already Ireland-specific (Europe/Dublin, EU waste carrier license)
FX required?	No	Yes — unless customer always prices/invoices in one currency

CS V3 evaluation — what to improve, not replicate

Why the CS V3 approach is unsuited to cloud international operations

CS V3 pattern	Problem	Cloud replacement
Currency implicit per company code	No explicit currency on quotes/orders; cross-currency	transactionCurrency on every monetary record from MVP

	requires duplicate price lists or manual spreadsheet conversion	
Pricing Company Code vs Shipping Company Code vs Shipping Plant Code	Three overlapping codes admins must understand and maintain; error-prone at border plants	Single Legal Entity + Location model; system resolves pricing/dispatch entity from context
Tax Code manually linked to customer	Admin must pre-configure every cross-border tax scenario; no address-driven resolution	Auto tax jurisdiction from seller entity + buyer country + delivery address + supply type
Separate modules (COMMANDquote, order entry, COMMANDreceivables)	Pricing/tax/currency context may not propagate cleanly	Unified quote-to-cash pipeline with locked monetary context at each stage transition
Per-site database / Citrix deployment	Configuration duplicated per install; no central tenant policy	Tenant-level policy : enabled currencies, FX source, default locale, regional compliance flags
ERP owns FX	Cloud has no rate snapshot; audit trail breaks between quote and invoice	Platform FX snapshot at quote acceptance; ERP receives immutable rate metadata
No presentation currency	NI salesperson cannot show EUR equivalent to ROI customer without manual calc	presentationCurrency optional field, auto-calculated from tenant FX feed

What to preserve from CS V3 (via import, not UX)

- Historical monetary amounts and tax codes exactly as stored (import fidelity per hauler spec)
- Company/plant/customer hierarchy **as import source fields**, mapped to Cloud canonical model
- Existing customer contracts that specify invoice currency

What CS V3 discovery must answer (Phase 0)

- Which CS V3 configurations are **workarounds** for missing multi-currency (dual price lists, manual FX) vs intentional business rules
- How much admin time is spent maintaining tax codes and company code permutations at border sites
- Which fields can be **** dropped from Cloud UI**** entirely and resolved automatically

CS V3 does not expose a first-class "multi-currency" module in available documentation. Instead, monetary values flow through a **company/plant/customer hierarchy** with **tax codes** determining jurisdiction.

```

flowchart LR
    subgraph masterData [Master Data]
        CompanyCode["Company Code"]
        PricingCo["Pricing Company Code"]
        ShippingCo["Shipping Company Code"]
    end

```

```

    PricingPlant["Pricing Plant Code"]
    ShippingPlant["Shipping Plant Code"]
    Customer["Customer + Tax Code"]
    DeliveryAddr["Delivery Address / Zone"]
end

subgraph lifecycle [Quote to Cash]
    Quote["COMMANDquote / Job"]
    Order["Order Entry"]
    Ticket["Ticket / Dispatch"]
    Invoice["COMMANDreceivables / Invoice"]
end

CompanyCode --> PricingCo
CompanyCode --> ShippingCo
PricingCo --> Quote
Customer --> Quote
DeliveryAddr --> Quote
Quote --> Order
PricingPlant --> Order
ShippingPlant --> Order
Customer --> Order
Order --> Ticket
Ticket --> Invoice
Customer --> Invoice

```

Currency determination (inferred)

Stage	Likely rule	User display
Quote	Prices resolved from Pricing Company Code + Pricing Plant Code + customer/job pricing tables. Currency is implicit — tied to the pricing company's legal entity / accounting setup , not the customer's country.	Amounts shown in the company's configured currency (EUR for EU deployments). No currency selector on quote screens in available docs.
Order	Inherits quote/project pricing. Shipping Plant Code may differ from Pricing Plant Code (common for border plants). Tax Code on customer/order drives VAT.	Same currency as pricing company. Tax code description visible via smart lookup.
Invoice	Posted from tickets/orders. Amount + tax breakdown. Cross-border B2B typically net + zero VAT (reverse charge / intracommunity supply) when both parties have valid EU VAT IDs — configured via tax codes, not currency conversion.	Invoice PDF/screen shows line amounts, tax code, net/gross per local accounting rules.

Key CS V3 configuration fields (from COMMANDexecutive user guide)

These fields appear on order/quote configuration and drive cross-border behaviour more than currency:

- **Pricing Company Code** — which legal entity's price list applies

- **Shipping Company Code** — which entity dispatches / owns the delivery
- **Pricing Plant Code / Shipping Plant Code** — plant-level pricing vs dispatch origin (relevant for a French plant serving a German site near the border)
- **Tax Code** — customer-linked; determines VAT treatment
- **Delivery Address / Zone Code** — delivery location for tax and freight
- **Accounting Category Code** — GL mapping downstream to ERP

France → Germany concrete scenario (EUR, cross-border)

For a French producer billing a German customer at a border plant:

1. **Quoting:** Sales creates quote under French **Pricing Company** price tables (EUR). Delivery address in Germany sets zone/freight; tax code reflects B2B cross-border rules.
2. **Orders: Shipping Plant** may be the French border plant; **Pricing Plant** may match or differ. Order inherits EUR amounts from quote.
3. **Invoicing:** Invoice issued in EUR from the French legal entity. VAT likely **0% with reverse charge** (both VAT-registered) or destination rules depending on goods vs services and movement — handled by **Tax Code**, not currency logic.
4. **Display:** Users see EUR amounts everywhere. Cross-border distinction is visible via customer country, delivery address, tax code description, and invoice legal text — not a currency switcher.

Northern Ireland / UK ↔ Republic of Ireland scenario (GBP/EUR, post-Brexit)

This corridor is especially relevant for International RMC: border plants near Derry/Londonderry, Newry, and Donegal routinely serve customers on both sides. The existing [hauler license spec](#) already targets **Ireland** (Europe/Dublin timezone, EU waste carrier compliance) — cross-border monetary support should extend the same regional pattern.

flowchart TB

```

subgraph entities [Legal Entities]
    NIPlant["NI Plant / Company GBP functional"]
    ROIPlant["ROI Plant / Company EUR functional"]
    GBPlant["GB Mainland Company GBP"]
end

subgraph customer [Cross-Border Customer]
    ROICust["ROI Customer EUR contract"]
    NICust["NI Customer GBP or EUR contract"]
end

subgraph rules [Tax and Currency Rules]
    Windsor["Windsor Framework goods = intra-EU VAT"]
    Services["Services = UK VAT rules"]
    FX["GBP/EUR rate snapshot at quote accept"]
end

NIPlant -->|"Quote in GBP or EUR per contract"| ROICust
ROIPlant -->|"Quote in EUR"| NICust
NIPlant --> Windsor
ROIPlant --> Windsor
NIPlant --> FX
ROIPlant --> FX

```

Direction A: NI producer → ROI customer (most common RMC pattern)

Example: Ready-mix plant in County Derry dispatching to a construction site in Donegal.

Stage	Currency	Tax / compliance
Quote	Typically GBP (NI company's functional currency) unless customer contract specifies EUR. If EUR requested, show both or convert with displayed FX rate.	Delivery address in ROI triggers intra-EU goods treatment. Customer IE VAT number required. Seller uses XI-prefixed VAT number on all EU-facing documentation.
Order	Locked to quote currency. Shipping Plant = NI border plant; Pricing Company = NI legal entity.	Tax Code reflects zero-rated B2B intra-EU supply of goods. Proof of movement may be required for audit.
Ticket / Dispatch	Amounts in transaction currency. Hauler EU license check applies (<code>isEuRegion = true</code> , Europe/Dublin).	Dispatch date evaluated in Europe/Dublin local calendar (per hauler spec).
Invoice	Invoice in quote currency (GBP or EUR). If ERP functional currency differs, export both amounts + FX rate snapshot .	Zero VAT on invoice; EC Sales List (ROI side) / intra-EU acquisition reporting. Invoice must show XI VAT number and customer IE VAT number.

Direction B: ROI producer → NI customer

Example: Aggregate supplier in Monaghan delivering to a site in Armagh.

Stage	Currency	Tax / compliance
Quote	EUR (ROI functional currency). NI customer may request GBP equivalent for their internal budgeting — display as secondary, not as legal quote currency unless contract dictates.	ROI seller charges no Irish VAT on B2B goods to NI (intra-EU supply).
Order / Invoice	EUR throughout.	NI buyer accounts for VAT via reverse charge / acquisition on their UK return.

Direction C: GB mainland → NI or ROI (different rules — not intra-EU)

Goods moving **Great Britain → NI** use Windsor Framework **green lane** (internal UK market, no EU customs). Goods moving **GB → ROI** are **exports** — full customs and export VAT rules apply, not the NI ↔ ROI intra-EU treatment. Command Cloud must **not conflate** GB ↔ ROI with NI ↔ ROI; the `sellerCountry` and `dispatchOriginCountry` fields must distinguish GB, NI (XI), and IE.

How users likely see this in CS V3 today (inferred)

- **Single currency per company code** — NI company shows **£**, ROI company shows **€**; no automatic conversion on screen unless manually maintained dual price lists.
- **Tax code** drives the cross-border distinction; users select or inherit a tax code mapped to "Intra-EU supply" or "Export" rather than choosing a currency.
- **Pricing Company Code** determines which entity's price list applies — critical when a group operates both an NI plant and an ROI plant under one database.

- **Gap:** If a NI salesperson quotes an ROI customer in EUR, CS V3 likely requires either a separate EUR price list or manual conversion outside the system — **this is the primary Command Cloud improvement opportunity.**

CS V3 gaps specific to NI/ROI (require validation)

- Whether CS V3 stores **XI-prefixed VAT numbers** distinctly from GB VAT numbers
- Whether any Irish/NI deployments maintain **dual price lists** (GBP and EUR) per customer or product
- How **EC Sales List / intra-EU acquisition** data is exported to Sage, SAP, or Xero
- Whether GB mainland, NI, and ROI are modelled as separate **Company Codes** or **Pricing Company Codes** within one tenant
- FX rate source (manual table, ERP feed, or none) for cross-currency quotes

CS V3 gaps / unknowns (require deeper investigation)

These cannot be confirmed without CS V3 source, DB schema, or live customer walkthroughs on **both** corridors:

- Whether any EU deployments store an explicit `currencyCode` per company/customer
- Whether CS V3 supports multiple currencies within one database (**especially relevant for NI/ROI**)
- Exact tax engine (built-in tables vs ERP delegation) for DE/FR VAT and NI/ROI Windsor Framework rules
- How CS V3 exports monetary fields to SAP/Sage/Viewpoint for cross-border GL posting
- UI localisation of number formatting (£1,234.56 vs 1.234,56 €) independent of currency
- Whether **XI VAT prefix** is stored and displayed correctly for NI entities

How Command Cloud Handles This Today

Command Cloud is moving toward a **centralised quote-to-cash** model with **location-based tax** and **ERP downstream** for full financial compliance.

flowchart TB

```

subgraph cloud [Command Cloud]
    SQ["Sales and Quoting"]
    Dispatch["Dispatch and Scale Ticketing"]
    Billing["Billing Prep / Invoicing / AR"]
    TaxSvc["Integrated Tax Calculation Service"]
    Payments["Payments via Fractal"]
end

subgraph apis [Command Cloud APIs]
    BillingAPI["Billing and Invoicing API"]
    ERPAPI["ERP and Reporting API"]
    PaymentsAPI["Payments API"]
end

subgraph external [External Systems]
    ERP["ERP / Accounting e.g. SAP Sage"]
    EInvoicing["E-Invoicing PA e.g. Peppol Factur-X"]
    VIES["EU VIES VAT Validation"]
end

SQ -->|"pricing flows through"| Dispatch

```

```

Dispatch --> Billing
Billing --> TaxSvc
Billing --> Payments
Billing --> BillingAPI
BillingAPI --> ERP
ERPAPI --> ERP
Billing --> EInvoicing
TaxSvc --> VIES

```

Current Command Cloud capabilities (public docs)

Capability	Status	Relevance to cross-border
Centralised pricing (job/quote/customer/location/company)	Existing	Same EUR pricing model as CS V3 hierarchy
Quote → dispatch → invoice price consistency	Existing	Prevents border mismatches
Integrated tax calculation (location + rules)	Existing	Handles DE/FR rate differences; provider not named publicly
Multi-legal-entity invoicing	Near-term roadmap	Critical for FR company / DE customer with separate books
Regional gating (<code>isEuRegion</code>)	Pattern established in hauler spec	Template for EU-only compliance features
Multilingual UI	Existing	Locale display; not currency
Explicit multi-currency / FX	Not documented	Critical gap for NI/ROI (GBP/EUR) ; likely delegated to ERP today
EU e-invoicing (Factur-X, XRechnung, Peppol)	Not documented	Major gap for FR/DE compliance
UK / ROI e-invoicing (2028–2029 mandates)	Not documented	Major gap for NI/ROI corridor
Ireland-specific compliance	Pattern in hauler spec	Europe/Dublin, EU waste carrier — extend to monetary/tax

Established migration pattern (from hauler spec)

The [hauler license spec](#) defines the **recommended Command Cloud approach** for EU cross-border features:

1. **Regional gate** at tenant/company level (`isEuRegion`)
2. **Canonical field contract** with ISO formats
3. **Dedicated REST endpoints** per domain object
4. **Import rules** — preserve CS V3 source values exactly; no silent coercion
5. **Server-side enforcement** with UI mirroring

6. Audit metadata on imports

This pattern should be reused for cross-border monetary/tax fields.

Recommended Command Cloud Target Architecture

MVP foundation — multi-currency data model (all tenants, all corridors)

Multi-currency is **not a Phase 3 feature**. It is the **platform baseline** from which corridor-specific compliance and advanced FX are layered.

Tenant setup (simplified admin — one-time)

Admins configure **once per tenant**, not per transaction:

Setting	Example	Purpose
operatingRegion	EU / UK / UK-NI	Enables regional compliance packs
enabledCurrencies	["EUR"] or ["GBP", "EUR"]	Which currencies can appear on quotes/invoices
defaultFxSource	ECB_DAILY	Platform-managed rate feed (no manual tables)
defaultLocale	en-IE	Number/date formatting default

Legal entity setup (per company/plant group)

Setting	Example	Purpose
functionalCurrency	GBP	Entity's accounting currency
vatRegistration	{ country: "GB-NI", number: "123456789", prefix: "XI" }	Tax identity — replaces scattered CS V3 tax/company codes
legalEntityCountry	GB-NI	Drives Windsor Framework eligibility

Customer setup (minimal — system infers the rest)

Setting	Example	Purpose
preferredInvoiceCurrency	EUR (optional)	Overrides default when contract requires
vatRegistration	{ country: "IE", number: "..." }	Validated via VIES/HMRC on save
billingCountry	IE	Used with delivery address for tax resolution

Not required in Cloud setup (inferred automatically):

- Pricing Company Code / Shipping Company Code — derived from legal entity + dispatch location
- Tax Code selection — resolved by tax jurisdiction service from address + supply type
- FX rate — fetched from tenant FX source unless contract-fixed rate applies
- Dual price lists per currency — single price list in functional currency; platform converts for presentation

Mandatory monetary fields (every quote, order, ticket line, invoice line — MVP)

```
{
  "amount": { "value": 10000.00, "currency": "GBP" },
  "functionalAmount": { "value": 10000.00, "currency": "GBP" },
  "transactionCurrency": "GBP",
  "functionalCurrency": "GBP",
  "presentationAmount": { "value": 11682.00, "currency": "EUR" },
  "presentationCurrency": "EUR",
  "exchangeRate": 1.1682,
  "exchangeRateSource": "ECB_DAILY",
  "exchangeRateDate": "2026-06-30",
  "exchangeRateLockedAt": null,
  "taxJurisdiction": {
    "sellerCountry": "GB-NI",
    "buyerCountry": "IE",
    "deliveryCountry": "IE",
    "supplyType": "GOODS",
    "vatTreatment": "INTRA_EU_B2B_ZERO_RATED",
    "resolvedBy": "TAX_JURISDICTION_SERVICE"
  }
}
```

When `transactionCurrency === functionalCurrency` (e.g. FR→DE both EUR), FX fields are present but null — **same schema, no special case.**

Auto-resolution at quote creation (replaces CS V3 manual configuration)

```
sequenceDiagram
    participant User
    participant Cloud as Command Cloud
    participant TaxSvc as Tax Jurisdiction Service
    participant FxSvc as FX Service

    User->>Cloud: Create quote for customer + delivery address
    Cloud->>Cloud: Resolve legal entity from user context + dispatch location
    Cloud->>Cloud: Set functionalCurrency from legal entity
    Cloud->>Cloud: Set transactionCurrency from customer.preferredInvoiceCurrency or functionalCurrency
    alt transactionCurrency differs from functionalCurrency
        Cloud->>FxSvc: Get rate for currency pair
        FxSvc-->>Cloud: Rate + source + date
        Cloud->>Cloud: Calculate presentationAmount
    end
    Cloud->>TaxSvc: Resolve jurisdiction seller + buyer + delivery + supplyType
    TaxSvc-->>Cloud: vatTreatment + compliance metadata
    Cloud-->>User: Quote with currency, tax badge, no manual code selection
```

Progressive enhancement ladder

Layer	Scope	Single-currency tenant (EUR only)	Multi-currency tenant (GBP/EUR)
MVP foundation	Data model + simplified setup + auto-resolution + locale display	Works: all amounts EUR, FX fields null	Works: dual display, FX snapshot at quote accept
Phase 1 — Corridor compliance	Tax rules packs, VIES/HMRC, e-invoicing PA	FR ↔ DE reverse charge, Peppol	NI ↔ ROI Windsor Framework, XI prefix
Phase 2 — Multi-entity	Intercompany pricing, consolidated AR	Multi-plant FR/DE entities	NI + ROI entities under one tenant
Phase 3 — Advanced FX	Contract-fixed rates, override approval, extra pairs (CHF/PLN), ERP revaluation export	Rarely needed	Contract EUR rates, finance override workflow

Data model extensions (examples)

Introduce explicit monetary context on quote, order, invoice, and line items.

Example A — France → Germany (EUR, same schema, FX null):

```
{
  "transactionCurrency": "EUR",
  "functionalCurrency": "EUR",
  "legalEntityId": "fr-legal-entity-001",
  "dispatchLocationId": "fr-plant-border-801",
  "taxJurisdiction": {
    "sellerCountry": "FR",
    "buyerCountry": "DE",
    "deliveryCountry": "DE",
    "vatTreatment": "INTRA_EU_B2B_REVERSE_CHARGE",
    "resolvedBy": "TAX_JURISDICTION_SERVICE"
  },
  "exchangeRate": null,
  "exchangeRateDate": null
}
```

Example B — NI producer → ROI customer (same schema, FX populated):

```
{
  "transactionCurrency": "GBP",
  "functionalCurrency": "GBP",
  "presentationCurrency": "EUR",
  "legalEntityId": "ni-legal-entity-004",
  "dispatchLocationId": "ni-plant-derry-012",
  "taxJurisdiction": {
    "sellerCountry": "GB-NI",
    "sellerVatPrefix": "XI",

```

```

    "buyerCountry": "IE",
    "deliveryCountry": "IE",
    "supplyType": "GOODS",
    "vatTreatment": "INTRA_EU_B2B_ZERO_RATED",
    "resolvedBy": "TAX_JURISDICTION_SERVICE"
  },
  "exchangeRate": 1.1682,
  "exchangeRateSource": "ECB_DAILY",
  "exchangeRateDate": "2026-06-30",
  "amount": { "value": 10000.00, "currency": "GBP" },
  "presentationAmount": { "value": 11682.00, "currency": "EUR" }
}

```

For multi-currency corridors (NI/ROI) and single-currency corridors (FR/DE) alike:

- **transactionCurrency and functionalCurrency are always stored** — never implicit
- Snapshot **FX rate at quote acceptance** when currencies differ; lock via `exchangeRateLockedAt`
- Never recalculate historical amounts when rates change
- ERP receives both transaction and functional amounts + rate metadata on every export

Quote → order → invoice currency rules (proposed)

Event	Rule
Quote creation	Auto-resolve transactionCurrency from customer preference → functional currency fallback; auto-resolve tax jurisdiction from addresses — no manual tax/currency code selection
Quote acceptance	Lock currency + FX rate snapshot when currencies differ
Order creation	Inherit locked monetary context; re-validate tax if delivery address changed
Ticketing	Amounts in transactionCurrency; no conversion at ticket time
Invoice	Same locked context; tax breakdown + e-invoice payload from taxJurisdiction

User display guidelines (MVP)

- **Default: show only what matters** — single-currency quotes show one amount; cross-currency quotes show primary + presentation with rate footnote
- Show **currency code + symbol** using `Intl.NumberFormat` with explicit `currency` and user locale
- When presentation currency differs: £10,000.00 (GBP) · approx. €11,682.00 @ ECB 30 Jun 2026
- Surface **tax treatment badge** resolved automatically — never ask user to interpret tax code numbers
- **Hide legacy CS V3 fields** from all Cloud UI (Pricing Company Code, etc.) — visible only in import mapping logs for migration support
- Distinguish **GB**, **GB-NI (XI)**, and **IE** in country selectors

Recommended API Integrations (Prioritised)

Tier 0 — MVP foundation (all tenants)

Integration	Purpose	Notes
Platform FX Service	Tenant-configured rate feed; snapshot + cache	ECB daily for EUR pairs; extensible to commercial feeds — built-in, not ERP-dependent
Tax Jurisdiction Service	Address-driven VAT treatment resolution	Replaces manual CS V3 tax code maintenance
Billing & Invoicing API (extend)	Multi-currency payload on every export from MVP	transactionCurrency, functionalAmount, exchangeRate, taxJurisdiction always present
ERP & Reporting API (extend)	GL handoff with rate metadata	ERP posts in functional currency; Cloud owns operational truth

Tier 1 — Corridor compliance packs (Phase 1)

Integration	Purpose	Notes
EU VIES VAT Validation API	Validate EU VAT IDs (FR, DE, IE)	Free; cache; runs on customer save
HMRC VAT Registration Check	Validate GB/NI VAT + XI prefix	Required for NI/ROI corridor pack
E-invoicing Access Point (Peppol / PA)	EN 16931 invoice delivery	B2Brouter, Storecove, or Pagero — one REST integration, corridor-specific output formats
Corridor rule packs	Pluggable tax logic	EU_CONTINENTAL (FR ↔ DE), UK_IRELAND_WINDSOR (NI ↔ ROI), UK_EXPORT (GB ↔ ROI)

Note: FX APIs are **Tier 0 (MVP)**, not deferred. Corridor packs add tax/compliance logic on top of the same multi-currency foundation.

Tier 2 — EU compliance and audit

Integration	Purpose
Peppol SMP / SML lookup	Resolve buyer's e-invoicing endpoint (Leitweg-ID for DE public sector)
France PPF / Flux 10 e-Reporting	Mandatory from Sep 2026 for FR entities — cross-border B2B reporting
Germany ZUGFeRD / XRechnung generation	B2B/B2G invoice formats for DE buyers
Audit trail API	Immutable log of tax decisions, VIES checks, rate snapshots

Tier 3 — Advanced FX and treasury (Phase 3)

Integration	Purpose
-------------	---------

Commercial FX APIs (Fixer.io, Open Exchange Rates)	Intraday rates, additional pairs (CHF, PLN, USD)
Contract-fixed rate override	Customer-specific locked rates independent of market feed
ERP revaluation export	Period-end FX revaluation entries for finance teams
Rate override approval workflow	Finance approves sales-team rate exceptions with audit trail

Phased Scoping Roadmap for Command Cloud

Phase 0 — CS V3 discovery (prerequisite)

Focus: understand CS V3 **workarounds to eliminate**, not features to replicate.

1. Authenticate Atlassian MCP; search for currency/cross-border specs and International RMC requirements
2. Open CS V3 codebase/schema — identify implicit currency storage and dual price list patterns
3. SME sessions on FR ↔ DE and NI/ROI — measure admin burden (tax codes, company codes, manual FX)
4. Produce **CS V3 → Cloud import mapping** and **"do not replicate" list** for UX

MVP foundation — multi-currency platform baseline (ship first)

Scope: Every tenant, every corridor. Single-currency EUR tenants use the same model with FX fields null.

Platform changes:

- Canonical monetary schema on quote, order, ticket line, invoice line (`transactionCurrency` , `functionalCurrency` , `amount` , `functionalAmount` , FX snapshot fields, `taxJurisdiction`)
- Simplified **Tenant Admin**: operating region, enabled currencies, FX source, default locale
- Simplified **Legal Entity Admin**: functional currency, VAT registration (replaces company code sprawl)
- Simplified **Customer Admin**: preferred invoice currency, VAT ID (validated async)
- **Tax Jurisdiction Service** v1: address-driven resolution; default domestic rules
- **Platform FX Service** v1: ECB daily rates for enabled pairs; snapshot at quote acceptance
- **Auto-resolution at quote creation** — no Pricing Company Code / Tax Code picker in UI
- Locale-aware currency display via `Intl.NumberFormat`
- Billing & Invoicing API exports full monetary context from day one
- CS V3 import: map legacy codes to canonical model; flag unmappable records for admin review

Single-currency FR/DE tenant experience: Admin sets `enabledCurrencies: ["EUR"]` . Quotes auto-resolve EUR/EUR, tax jurisdiction still auto-resolved. Zero extra admin vs today.

Multi-currency NI/ROI tenant experience: Admin sets `enabledCurrencies: ["GBP", "EUR"]` , configures NI legal entity with `functionalCurrency: GBP` . Quotes to ROI customers auto-show EUR presentation. No dual price lists.

Phase 1 — Corridor compliance packs (on MVP foundation)

Pluggable rule packs activated by tenant region + corridor — **not separate products**.

EU Continental pack (FR ↔ DE):

- Intra-EU B2B reverse charge / zero-rated goods rules
- VIES validation for FR/DE VAT IDs
- E-invoicing: Peppol BIS 3.0 + Factur-X/PPF for FR statutory reporting

UK-Ireland pack (NI ↔ ROI):

- Windsor Framework rules engine (goods vs services; NI ↔ ROI vs GB ↔ ROI)
- XI VAT prefix handling; HMRC + VIES validation
- E-invoicing: Peppol-ready for ROI Nov 2028 and UK Apr 2029 mandates

CS V3 migration: Import legacy tax codes as audit reference only; Cloud resolves jurisdiction afresh from addresses post-migration.

Phase 2 — Multi-legal-entity billing

- Pricing entity vs invoicing entity vs dispatch entity (without exposing CS V3 code names)
- Consolidated customer view with entity-specific AR
- Intercompany markup rules for border plants

Phase 3 — Advanced FX and treasury

For tenants needing more than MVP FX (not required for launch):

- Contract-fixed exchange rates per customer
- Finance approval workflow for rate overrides
- Additional currency pairs beyond tenant `enabledCurrencies` defaults
- ERP period-end revaluation export
- Intraday commercial rate feeds

Best Practices to Adopt

1. **Multi-currency in the data model from MVP** — even EUR-only tenants; avoids a costly migration later and eliminates CS V3 implicit currency
2. **Simplify admin setup ruthlessly** — if a CS V3 field requires a consultant to explain it, it does not belong in Cloud UI
3. **Infer, don't configure** — tax jurisdiction and default currency from address + entity + customer contract; reserve manual override for exceptions only
4. **Separate currency from tax jurisdiction** — FR ↔ DE is tax-first; NI ↔ ROI is tax + currency; same platform handles both via one schema
5. **Never conflate GB, GB-NI, and IE** — Windsor Framework applies to NI for goods only
6. **Snapshot rates at quote acceptance** — platform-owned, not ERP-delegated; audit trail in Cloud
7. **ERP remains statutory system of record** — Cloud exports immutable monetary context; ERP posts GL and VAT returns
8. **Import fidelity, UX modernisation** — preserve CS V3 amounts on migration; do not preserve CS V3 configuration UX
9. **Corridor compliance as pluggable packs** — not per-customer bespoke configuration
10. **Progressive enhancement** — MVP FX (ECB daily) is sufficient for NI/ROI launch; advanced treasury is Phase 3

Key Risks and Dependencies

Risk	Mitigation
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CS V3 behaviour varies by customer configuration	Phase 0 customer/SME validation on both corridors before API design
France e-invoicing mandate (Sep 2026)	Prioritise accredited PA integration in Phase 1a
ROI e-invoicing (Nov 2028) and UK mandate (Apr 2029)	Phase 1b Peppol-first architecture; monitor HMRC Budget 2026 roadmap
Windsor Framework rule changes	Abstract tax rules into configurable country-pair engine; avoid hard-coding
NI XI VAT prefix not supported in CS V3 or ERP exports	Explicit sellerVatPrefix field; validate in Phase 0
Multi-legal-entity not yet GA in Cloud	Phase 2 may depend on platform roadmap item
Replicating CS V3 setup complexity in Cloud	Explicit "do not replicate" list; UX review gate before Phase 1
Single-currency tenants forced through FX UI	MVP schema with null FX fields; UI adapts to single-currency context
MVP delayed waiting for Phase 3 advanced FX	Ship MVP foundation first; ECB daily rates sufficient for NI/ROI
No CS V3 source in current workspace	Blocker for code-level confirmation — needs repo access
Internal specs inaccessible	Authenticate Atlassian MCP or request Confluence export

Recommended Next Actions

1. **Draft MVP monetary schema** — mandatory currency fields on all transactional entities; review with platform architecture team before corridor packs
2. **Draft simplified Tenant/Legal Entity/Customer admin spec** — replace CS V3 company code / tax code setup with 3-screen cloud config
3. **Phase 0 CS V3 discovery** — focus on workarounds to eliminate; authenticate Atlassian; SME walkthroughs FR ↔ DE and NI/ROI
4. **Define corridor compliance packs** as pluggable modules on MVP foundation (EU Continental, UK-Ireland Windsor)
5. **Evaluate e-invoicing PA vendors** for single REST integration (Peppol-first, Factur-X/XRechnung output)
6. **Extend Billing & Invoicing API** with full monetary context from MVP — not as a later add-on
7. **CS V3 import mapping doc** — legacy field → canonical model; legacy fields hidden from Cloud UI post-migration